PHYS 365 — Final Exam Fall 2025 — Worked Solutions

25 questions. Every step uses the formula sheet printed at the back of the F2025 final. Anything that is *not* on that sheet is put in an orange **REMEMBER** box. Every final answer was checked against the official answer key.

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	B	6	A	11	A	16	B	21	C
2	A	7	D	12	B	17	C	22	C
3	C	8	C	13	A	18	B	23	A
4	C	9	D	14	B	19	A	24	D
5	A	10	D	15	C	20	B	25	D

REMEMBER — not on the formula sheet

- Wave speed is set by the medium. Frequency is set by the source (Q1).
- When two close tones play together you hear their **average** frequency, beating at $f_1 - f_2$ (Q5).
- SHM velocity: $v = -\omega x_m \sin(\omega t + \phi)$. Use its sign to pick between the two solutions of $\cos^{-1}$ (Q6).
- Phasor addition: add the x and y components, then $A = \sqrt{A_x^2 + A_y^2},\ \phi = \tan^{-1}\left(\dfrac{A_y}{A_x}\right)$ (Q7).
- Small angles: $y = D\tan\theta \approx D\sin\theta$ (Q11).
- Zero reflection: $Z_1 = Z_2$ (Q13).
- Missing double-slit order: $\dfrac{d}{a} = m_{\text{missing}}$ (Q15).
- Momentum $p = mv$, so $K = \dfrac{1}{2}mv^2 = \dfrac{p^2}{2m}$ (Q16, Q25).
- Photon energy from wavelength: $E = hf = \dfrac{hc}{\lambda}$ (combine $E = hf$ with $c = f\lambda$). Handy: $hc = 1240\ \text{eV}\cdot\text{nm}$ (Q20, Q23).
- "Second excited state" means $n = 3$. "Third excited state" means $n = 4$. The ground state is $n = 1$ (Q17, Q22).
- Photon momentum uncertainty: take $\Delta p \approx p = \dfrac{h}{\lambda}$ (Q19).
- Compton scattering: the m in $\Delta\lambda$ is the **electron** mass, because photons scatter off the target's electrons (Q23).

Question 1 What sets wave speed (FALSE statement)

1. A source is emitting acoustic waves in a particular medium. Which of the following statements is FALSE?

   A) The speed of the waves is determined by the elastic and inertial properties of the medium.
   B) The speed of the waves is determined by the frequency of the source.
   C) The velocity of the particles of the medium changes with time as the wave passes by.
   D) The wavenumber depends on the medium while the angular frequency does not.

FROM THE FORMULA SHEET

$$v = \sqrt{\frac{B}{\rho}}, \quad v = \sqrt{\frac{\gamma RT}{M}}, \quad v = f\lambda$$

Step 1. The speed depends only on the medium's elasticity (B) and inertia (ρ), so A is true.

Step 2. Changing the source frequency changes $\lambda = \frac{v}{f}$, not v. So "speed is determined by the frequency" is **false**.

Step 3. C is true: particle velocity oscillates. D is true: ω is fixed by the source, and $k = \frac{\omega}{v}$ depends on the medium.

Final answer: **(B) Speed is determined by the source frequency (false)** ✓ Matches answer key

Question 2 Which waves have the same speed?

2. Which of the following traveling waves have the same speed?

   I. $D(x,t) = 1.2\ \cos(3x - 2t)$
   II. $D(y,t) = 3.2\ \sin\left(6y + 4t + \frac{\pi}{4}\right)$
   III. $D(y,t) = 3.2\ \sin\left(2y - 4t + \frac{\pi}{4}\right)$
   IV. $D(x,t) = 3.2\ \sin\left(2x + 3t + \frac{\pi}{2}\right)$

   A) I and II
   B) II and III
   C) I and IV
   D) III and IV

FROM THE FORMULA SHEET

$v = \dfrac{\omega}{k}$

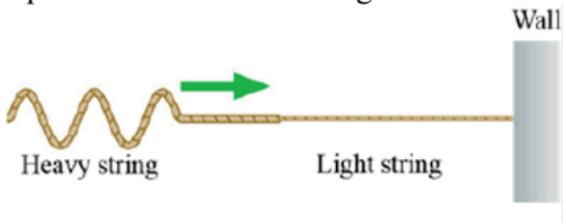
$$\text{I: } \frac{2}{3} \qquad \text{II: } \frac{4}{6} = \frac{2}{3} \qquad \text{III: } \frac{4}{2} = 2 \qquad \text{IV: } \frac{3}{2}$$

I and II both travel at $\frac{2}{3}$ (the direction and phase constants don't affect the speed).

Final answer: **(A) I and II** ✓ Matches answer key

Question 3 Impedance of joined strings

3. Two strings, one heavy and one light, are joined together and the light string is attached to the wall. A travelling wave is sent along the heavy string as shown. The linear mass density of the heavy string is twice as large as the linear mass density of the light string. Compare the mechanical impedances of the two strings.



A) $Z_{heavy} = \frac{Z_{light}}{2}$
B) $Z_{heavy} = \frac{Z_{light}}{\sqrt{2}}$
C) $Z_{heavy} = \sqrt{2} Z_{light}$
D) $Z_{heavy} = 2Z_{light}$

FROM THE FORMULA SHEET

$Z = \sqrt{\mu\tau}$

Step 1. The strings are joined, so they have the same tension. With $\mu_{heavy} = 2\mu_{light}$:

$$\frac{Z_{heavy}}{Z_{light}} = \sqrt{\frac{2\mu_{light}\,\tau}{\mu_{light}\,\tau}} = \sqrt{2} \;\Rightarrow\; Z_{heavy} = \sqrt{2}\, Z_{light}$$

Final answer: **(C)** $Z_{heavy} = \sqrt{2}\, Z_{light}$ ✓ Matches answer key

Question 4 Pendulum on Mars

4. A grandfather clock uses a pendulum with a period on Earth $T_{Earth} = 2.00$ s. If this clock were placed on Mars, where the gravitational acceleration is $g_{Mars} = 3.728\ \text{m/s}^2$, by how much would the length of the pendulum need to shorten to preserve its period? The gravitational acceleration of Earth is $g_{Earth} = 9.807\ \text{m/s}^2$

   A) $\Delta L = 0.994$ m

   B) $\Delta L = 0.378$ m

   C) $\Delta L = 0.616$ m

   D) $\Delta L = 0.497$ m

FROM THE FORMULA SHEET

$$T = 2\pi\sqrt{\frac{\ell}{g}}$$

Step 1. Solve for the length: $\ell = \dfrac{gT^2}{4\pi^2}$

$$\ell_E = \frac{(9.807)(2.00)^2}{4\pi^2} = 0.9937\ \text{m}, \qquad \ell_M = \frac{(3.728)(2.00)^2}{4\pi^2} = 0.3777\ \text{m}$$

Step 2. Change in length: $\Delta L = 0.9937 - 0.3777 = 0.616$ m

Final answer: **(C)** $\Delta L = 0.616$ **m** ✓ **Matches answer key**

Question 5 Two bells, beats, and Doppler

5. A cyclist has two slightly different bells mounted to the handlebars of their bike. They ring both bells at the same time as they travel towards a stationary microphone at 8 m/s. The cyclist hears a tone with a frequency of 257 Hz and 10 beats/s. A stationary microphone is positioned ahead of the cyclist. What frequency does the mic register? Use 340 m/s for speed of sound.

   A) The microphone will register a frequency of 263 Hz, beating with a frequency of 10.24 Hz
   B) The microphone will register a frequency of 251 Hz, beating with a frequency of 9.77 Hz
   C) The microphone will register a frequency of 263 Hz, beating with a frequency of 10 Hz
   D) The microphone will register a frequency of 251 Hz, beating with a frequency of 10 Hz

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FROM THE FORMULA SHEET

$$f' = f\,\frac{v \pm v_D}{v \pm v_S}, \qquad f_{beat} = f_1 - f_2$$

REMEMBER — not on the formula sheet

Two close tones are heard at their average frequency, $\dfrac{f_1 + f_2}{2}$, beating at $f_1 - f_2$.

Step 1. The cyclist moves with the bells, so they hear the true values: average tone 257 Hz and $f_1 - f_2 = 10$ Hz, which gives $f_1 = 262$ Hz and $f_2 = 252$ Hz.

Step 2. The source approaches a stationary microphone, so every frequency is multiplied by the same factor:

$$\frac{v}{v - v_S} = \frac{340}{340 - 8} = \frac{340}{332} = 1.0241$$

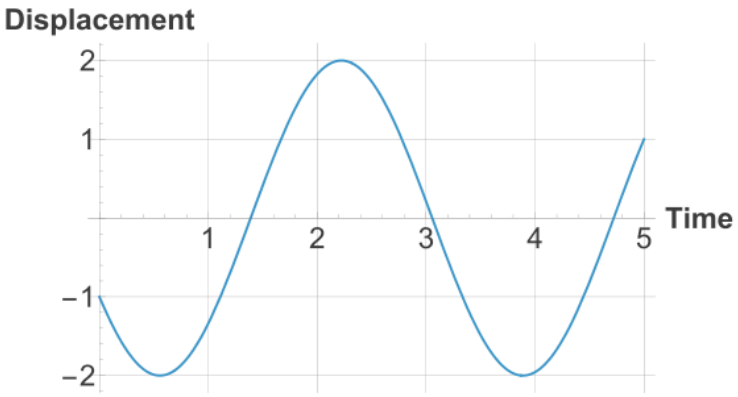
Step 3. Tone: $257 \times 1.0241 = 263$ Hz. Beat: $(262 - 252) \times 1.0241 = 10.24$ Hz.

Final answer: **(A) 263 Hz, beating at 10.24 Hz** ✓ Matches answer key

Question 6 Phase constant from a graph

6. What is the phase angle of a Simple Harmonic Motion graphically depicted by the image?

A) $\phi = \frac{2\pi}{3}$
B) $\phi = -\frac{\pi}{3}$
C) $\phi = \frac{\pi}{4}$
D) $\phi = 0$



FROM THE FORMULA SHEET

$$x(t) = x_m \cos(\omega t + \phi)$$

REMEMBER — not on the formula sheet

$$v(t) = -\omega x_m \sin(\omega t + \phi)$$

Step 1. From the graph, $x_m = 2$ and $x(0) = -1$:

$$-1 = 2\cos\phi \;\Rightarrow\; \cos\phi = -\frac{1}{2} \;\Rightarrow\; \phi = \pm\frac{2\pi}{3}$$

Step 2. At $t = 0$ the curve is going **down**, so $v(0) < 0$. That means $-\omega x_m \sin\phi < 0$, so $\sin\phi > 0$. Therefore $\phi = +\dfrac{2\pi}{3}$.

Final answer: **(A)** $\phi = \frac{2\pi}{3}$ ✓ **Matches answer key**

Question 7 Adding three SHMs with phasors

7. What are the amplitude and phase angle of a superposition of the following 3 SHMs?

$$y_1(t) = 2\cos(\omega t + \pi/2)$$
$$y_2(t) = 2\cos(\omega t)$$
$$y_3(t) = 3\cos(\omega t + \pi/4)$$

A) $A = 3$ and $\phi = 0$
B) $A = 7$ and $\phi = 3\pi/4$
C) $A = 2.83$ and $\phi = \pi$
D) $A = 5.83$ and $\phi = \pi/4$

FROM THE FORMULA SHEET
$\tilde{A} = Ae^{j\theta} = A(\cos\theta + j\sin\theta)$

Step 1. Components:

$$\tilde{y}_1 = 2\angle\tfrac{\pi}{2} = 0 + 2j, \qquad \tilde{y}_2 = 2\angle 0 = 2 + 0j, \qquad \tilde{y}_3 = 3\angle\tfrac{\pi}{4} = 2.121 + 2.121j$$

Step 2. Sum: $\tilde{y} = 4.121 + 4.121j$

Step 3. Amplitude and phase:

$$A = \sqrt{4.121^2 + 4.121^2} = 4.121\sqrt{2} = 5.83, \qquad \phi = \tan^{-1}\left(\frac{4.121}{4.121}\right) = \frac{\pi}{4}$$

Final answer: **(D)** $A = 5.83$, $\phi = \frac{\pi}{4}$ ✓ **Matches answer key**

Question 8 String driving a resonating tube

8. A nylon guitar string is fixed between two lab posts 2.00 m apart. The string has a linear mass density of $\mu = 7.20$ g/m and is placed under a tension of 160 N. The string is placed next to a tube, open at both ends, of length L. The string is plucked and resonate at the fundamental frequency while the tube resonates at the $n = 3$ mode. The speed of sound is 343 m/s. What is the length of the tube?

A) 4.63 m
B) 9.21 m
C) 13.8 m
D) 37.3 m

FROM THE FORMULA SHEET

$v = \sqrt{\dfrac{\tau}{\mu}}, \quad f_n = \dfrac{nv}{2L}$ (taut string or open–open pipe)

Step 1. String wave speed and fundamental ($n = 1$, $L_s = 2.00$ m):

$$v_s = \sqrt{\frac{160}{7.20 \times 10^{-3}}} = \sqrt{22222} = 149.1 \ \tfrac{\text{m}}{\text{s}}, \qquad f_1 = \frac{v_s}{2L_s} = \frac{149.1}{4.00} = 37.27 \text{ Hz}$$

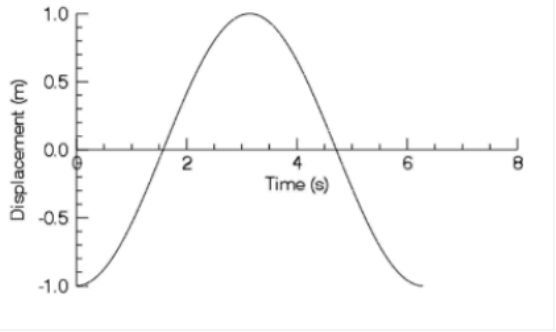
Step 2. The tube resonates at the same frequency in its $n = 3$ mode:

$$37.27 = \frac{3(343)}{2L} \ \Rightarrow \ L = \frac{3(343)}{2(37.27)} = \frac{1029}{74.54} = 13.8 \text{ m}$$

Final answer: **(C) 13.8 m** ✓ Matches answer key

Question 9 Standing wave from $y(t)$ at an antinode

9. A standing wave with a node at $x = 0$ is shown. The figure illustrates the displacement as a function of time at the first antinode $\left(x = \frac{\lambda}{4}\right)$. Which of the following equations correctly describes this standing wave?



A) $F(x,t) = 1.0\cos(kx)\cos(\omega t)$
B) $F(x,t) = -1.0\cos(kx)\sin(\omega t)$
C) $F(x,t) = 1.0\sin(kx)\cos(\omega t)$
D) $F(x,t) = -1.0\sin(kx)\cos(\omega t)$

FROM THE FORMULA SHEET
$y'(x,t) = [2y_m \sin(kx)]\cos(\omega t)$

Step 1. A node at $x = 0$ requires $\sin(kx)$, since $\sin 0 = 0$. That rules out A and B.

Step 2. At the antinode $x = \frac{\lambda}{4}$: $kx = \frac{2\pi}{\lambda} \cdot \frac{\lambda}{4} = \frac{\pi}{2}$, so $\sin(kx) = 1$ and $F = (\text{amplitude})\cos(\omega t)$.

Step 3. The graph starts at -1.0 at $t = 0$, so the amplitude coefficient is -1.0:

$$F(x,t) = -1.0\sin(kx)\cos(\omega t)$$

Final answer: **(D)** $F = -1.0\sin(kx)\cos(\omega t)$ ✓ **Matches answer key**

Question 10 Reading f and λ from a wave equation

10. A travelling wave is described by the equation $F(x,t) = 60\cos\big(3\pi(6t + x)\big)$. What are the frequency f and the wavelength λ of the wave?

A) 10 Hz and 2 m
B) 2 Hz and10 m
C) 0.67 Hz and 9 m
D) 9 Hz and 0.67 m

FROM THE FORMULA SHEET

$\omega = 2\pi f, \quad k = \dfrac{2\pi}{\lambda}$

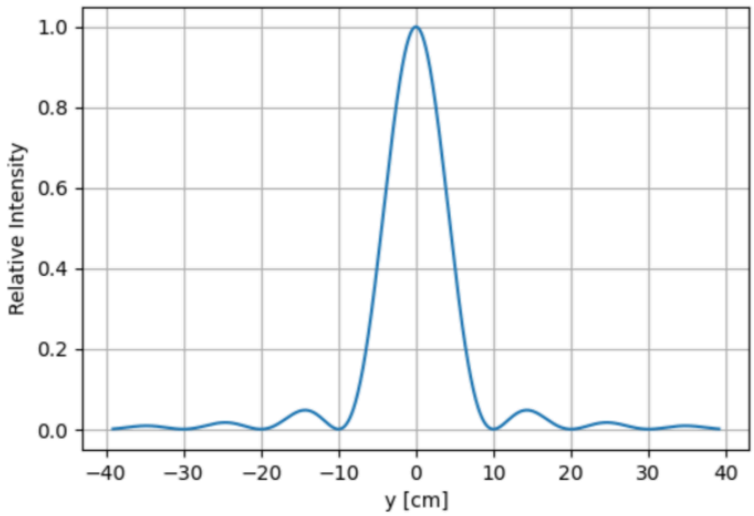
Step 1. Expand: $3\pi(6t + x) = 18\pi t + 3\pi x$, so $\omega = 18\pi$ and $k = 3\pi$.

$$f = \frac{18\pi}{2\pi} = 9 \text{ Hz}, \qquad \lambda = \frac{2\pi}{3\pi} = \frac{2}{3} = 0.67 \text{ m}$$

Final answer: **(D) 9 Hz and 0.67 m** ✓ Matches answer key

Question 11 Slit width from the diffraction graph

11. Light of 450 nm wavelength is shone through a slit to a screen 3.2 m beyond the slit. Based on the diffraction pattern in the graph below, what is the width of the slit?



A) 14.4 µm
B) 20 cm
C) 7.2 µm
D) 4.5 µm

FROM THE FORMULA SHEET

$a \sin\theta = m\lambda$ (minima)

REMEMBER — not on the formula sheet

Small angles: $\sin\theta \approx \tan\theta = \dfrac{y}{D}$

Step 1. From the graph, the first minima ($m = 1$) are at $y = \pm 10$ cm $= 0.10$ m.

Step 2. Solve for a:

$$a = \frac{\lambda}{\sin\theta} = \frac{\lambda D}{y} = \frac{(450 \times 10^{-9})(3.2)}{0.10} = 1.44 \times 10^{-5} \text{ m} = 14.4\,\mu\text{m}$$

Final answer: **(A) 14.4 µm** ✓ Matches answer key

Question 12 Power carried by a string wave

12. A string of length 5 m and mass 90 g is under a tension of 100 N. A wave travels along the string and is modeled by the function $y(x,t) = 0.01\sin(15.7x - 1170.12t)$.What is the power transmitted over one wavelength?

A) 7.3 W
B) 91.8 W
C) 9.5 kW
D) 0.12 kW

FROM THE FORMULA SHEET

$$v = \sqrt{\frac{\tau}{\mu}}, \quad P_{avg} = \frac{1}{2}\mu v\omega^2 y_m^2$$

REMEMBER — not on the formula sheet

$\mu = \dfrac{m}{L}$ (mass per unit length)

Step 1. Linear density and speed:

$$\mu = \frac{0.090}{5} = 0.018\ \tfrac{\text{kg}}{\text{m}}, \qquad v = \sqrt{\frac{100}{0.018}} = 74.5\ \tfrac{\text{m}}{\text{s}}$$

Check: $\dfrac{\omega}{k} = \dfrac{1170.12}{15.7} = 74.5$ ✓

Step 2. Average power, with $y_m = 0.01$ m and $\omega = 1170.12\ \tfrac{\text{rad}}{\text{s}}$:

$$P = \frac{1}{2}(0.018)(74.5)(1170.12)^2(0.01)^2 = 91.8\ \text{W}$$

Final answer: **(B) 91.8 W** ✓ Matches answer key

Question 13 Non-reflecting foam

13. An acoustical foam has a density of 29 kg/m^3. What bulk modulus should it have in order to reflect no sound incident normally on its surface? Assume the following properties for air: $B_{air} = 1.42 \times 10^5$ Pa and $\rho_{air} = 1.2\ \text{kg/m}^3$.

A) 5.88×10^3 Pa
B) 1.42×10^5 Pa
C) 1.20×10^3 Pa
D) 3.00×10^4 Pa

FROM THE FORMULA SHEET

$$R = \frac{Z_1 - Z_2}{Z_1 + Z_2}, \quad Z = \sqrt{\rho B}$$

Step 1. $R = 0$ requires $Z_{air} = Z_{foam}$:

$$\rho_{air} B_{air} = \rho_f B_f \;\Rightarrow\; B_f = \frac{(1.2)(1.42 \times 10^5)}{29} = 5.88 \times 10^3 \text{ Pa}$$

Final answer: **(A)** 5.88×10^3 **Pa** ✓ Matches answer key

Question 14 Reflection phase shifts in a soap film

14. A soap bubble is floating in air. Soap bubbles are a thin film of soap ($n = 1.35$) surrounding air ($n = 1.0$). When light hits the surface of a soap bubble with normal incidence, what is the phase difference **introduced by reflection**?

A) 0
B) π
C) 2π
D) $\pi/2$

FROM THE FORMULA SHEET

Reflection off a lower index gives 0. Off a higher index gives 0.5 wavelength.

Step 1. Front surface (air $\rightarrow$ soap, off a **higher** index): shift $\frac{1}{2}\lambda$, which is π.

Step 2. Back surface (soap $\rightarrow$ air, off a **lower** index): shift 0.

Step 3. Difference introduced by reflection: $\pi - 0 = \pi$

Final answer: **(B)** π ✓ Matches answer key

Question 15 Missing order gives slit separation

15. You are performing a double slit-experiment using a laser ($\lambda = 400$ nm) and two slits (each 2.4 μm wide). You do not know the distance between the slits but notice that the $m = 4$ bright fringe is missing, and coincides with the first diffraction minimum. What is the distance between the slits?

A) 1.6 μm
B) 7.2 μm
C) 9.6 μm
D) 19.2 nm

FROM THE FORMULA SHEET
$d \sin\theta = m\lambda$ (bright), $a \sin\theta = m\lambda$ (diffraction minima)

Step 1. At the same angle, $d \sin\theta = 4\lambda$ and $a \sin\theta = 1\lambda$. Divide:

$$\frac{d}{a} = 4 \;\Rightarrow\; d = 4(2.4\,\mu\text{m}) = 9.6\,\mu\text{m}$$

Final answer: **(C) 9.6 µm** ✓ Matches answer key

Question 16 Same de Broglie wavelength

16. Consider an electron ($m_e = 9.11 \times 10^{-31}$ kg) that has the same de Broglie wavelength as a neutron ($m_n = 1.67 \times 10^{-27}$ kg). Which of the following statements is true?

A) The neutron would have greater kinetic energy, speed, and momentum than the electron
B) The neutron and electron would have the same momentum, but the electron would have a greater speed and kinetic energy.
C) The neutron and electron would have the same momentum but the neutron would have a greater speed and kinetic energy
D) The electron would have a greater speed, kinetic energy, and momentum than the neutron.

FROM THE FORMULA SHEET
$p = \dfrac{h}{\lambda}$, $K = \dfrac{1}{2}mv^2$

REMEMBER — not on the formula sheet
$p = mv$, so $K = \dfrac{p^2}{2m}$.

Step 1. Same λ gives the same $p = \dfrac{h}{\lambda}$.

Step 2. With equal p: $v = \dfrac{p}{m}$ and $K = \dfrac{p^2}{2m}$. The electron's much smaller mass gives it a larger speed and a larger kinetic energy.

Final answer: **(B) Same momentum; electron has greater speed and KE** ✓ Matches answer key

Question 17 Longest wavelength emission from $n = 3$

17. A hydrogen atom is in its second excited state. Which energy level should it transition to in order to emit light of the longest possible wavelength?

A) $n = 0$
B) $n = 1$
C) $n = 2$
D) $n = 3$

FROM THE FORMULA SHEET

$$E_n = \frac{-13.6\ \text{eV}}{n^2}, \quad E = hf$$

Step 1. The second excited state is $n = 3$. A longer wavelength means a smaller photon energy, so we want the smallest drop.

$$3 \to 2: \ 13.6\left(\frac{1}{4} - \frac{1}{9}\right) = 1.89\ \text{eV} \qquad 3 \to 1: \ 13.6\left(1 - \frac{1}{9}\right) = 12.1\ \text{eV}$$

So the transition to $n = 2$ gives the longest wavelength. ($n = 0$ does not exist.)

Final answer: **(C)** $n = 2$ ✓ **Matches answer key**

Question 18 Width of an infinite well

18. An electron is confined in a one-dimensional infinite potential well. If the energy of the electron at the $n = 8$ energy level is measured to be 14.10 eV, what is the width L of the well?

A) 0.42 nm
B) 1.31 nm
C) 2.62 nm
D) 10.48 nm

FROM THE FORMULA SHEET

$$E_n = \frac{h^2}{8m_eL^2}n^2, \quad e = 1.602 \times 10^{-19}\ \text{C}$$

Step 1. Convert the energy: $E_8 = 14.10\ \text{eV} = 14.10(1.602 \times 10^{-19}) = 2.259 \times 10^{-18}$ J

Step 2. Solve for L:

$$L = \frac{nh}{\sqrt{8m_e E_n}} = \frac{8(6.63 \times 10^{-34})}{\sqrt{8(9.109 \times 10^{-31})(2.259 \times 10^{-18})}} = \frac{5.304 \times 10^{-33}}{4.057 \times 10^{-24}} = 1.31 \times 10^{-9}\ \text{m}$$

Final answer: **(B) 1.31 nm** ✓ Matches answer key

Question 19 Heisenberg limit on position

19. In microscopy, the Heisenberg uncertainty principle limits how precisely atomic positions can be measured. When a sample is observed, photons scatter off the atoms, perturbing their motion. A photon with wavelength $\lambda = 440$ nm scatters off an atom in a crystalline lattice. Using the uncertainty principle, determine the minimum uncertainty in the atom's position.

A) 35 nm
B) 75 nm
C) 110 nm
D) 220 nm

FROM THE FORMULA SHEET

$$\Delta x\, \Delta p_x \geq \frac{h}{4\pi}, \quad p = \frac{h}{\lambda}$$

REMEMBER — not on the formula sheet

The scattered photon can transfer up to its whole momentum, so $\Delta p \approx p_{photon} = \frac{h}{\lambda}$.

$$\Delta x \geq \frac{h}{4\pi\, \Delta p} = \frac{h}{4\pi \cdot \frac{h}{\lambda}} = \frac{\lambda}{4\pi} = \frac{440\ \text{nm}}{4\pi} = 35\ \text{nm}$$

Final answer: **(A) 35 nm** ✓ Matches answer key

Question 20 Photoelectric threshold

20. You have two sources of radiation: Source A has a wavelength 520 nm, and Source B has a wavelength of 300 nm. Source A emits light that is twice the intensity of Source B. You direct both sources at a sheet Aluminum ($\Phi = 4.08$ eV). You measure the stopping voltage and photocurrent in both cases. Which of the following statements is true?

A) The stopping voltage will be lower for Source A, but the photocurrent will be higher
B) A photocurrent and stopping voltage will only be measurable for source B
C) The stopping voltage will be higher for source A
D) The photocurrent will be higher for source A

FROM THE FORMULA SHEET

$E = hf, \quad hf = K_{max} + \Phi, \quad h = 4.136 \times 10^{-15}\ \text{eV} \cdot \text{s}$

REMEMBER — not on the formula sheet

$E = \dfrac{hc}{\lambda}$, with $hc = (4.136 \times 10^{-15})(3.00 \times 10^{8}) = 1.24 \times 10^{-6}\ \text{eV} \cdot \text{m} = 1240\ \text{eV} \cdot \text{nm}$

Step 1. Photon energies:

$$E_A = \frac{1240}{520} = 2.38\ \text{eV} < 4.08\ \text{eV}, \qquad E_B = \frac{1240}{300} = 4.13\ \text{eV} > 4.08\ \text{eV}$$

Step 2. Source A cannot eject electrons at **any** intensity. Source B gives $K_{max} = 0.05$ eV, so a current and a stopping voltage exist only for B.

Final answer: **(B) Only source B gives a photocurrent and stopping voltage** ✓ Matches answer key

Question 21 Most likely positions for $n = 3$

21. An electron confined to a 1-dimensional infinite potential well is in the $n = 3$ energy state. Where is it most likely to be found?

A) At the center of the box
B) One-quarter and three-quarters of the way from the ends
C) One-sixth, one-half, and five-sixths of the way from the ends
D) Equally likely to be found anywhere in the box

FROM THE FORMULA SHEET

$\psi_n(x) = \sqrt{\dfrac{2}{L}} \sin\left(\dfrac{n\pi}{L}x\right)$

Step 1. $|\psi_3|^2 \propto \sin^2\left(\dfrac{3\pi x}{L}\right)$ is at its maximum where $\dfrac{3\pi x}{L} = \dfrac{\pi}{2}, \dfrac{3\pi}{2}, \dfrac{5\pi}{2}$:

$$x = \frac{L}{6}, \ \frac{L}{2}, \ \frac{5L}{6}$$

Final answer: **(C) $\frac{1}{6}$, $\frac{1}{2}$, $\frac{5}{6}$ of the way** ✓ Matches answer key

Question 22 Probability in the leftmost eighth

22. An electron is confined in a one-dimensional infinite potential well of width $L = 200$ pm. The electron is in the third excited state. What is the probability of finding the electron in the leftmost eighth of the well.

A) 0.091
B) 0.91
C) 0.125
D) 0.25

FROM THE FORMULA SHEET

$$P(x_1, x_2) = \int_{x_1}^{x_2} |\psi(x)|^2 dx, \qquad \int \sin^2(ax)\, dx = \frac{x}{2} - \frac{\sin(2ax)}{4a} + c$$

Step 1. The third excited state is $n = 4$, so $a = \dfrac{4\pi}{L}$:

$$P = \frac{2}{L}\int_0^{L/8} \sin^2\left(\frac{4\pi x}{L}\right) dx = \frac{2}{L}\left[\frac{x}{2} - \frac{L\sin\left(\frac{8\pi x}{L}\right)}{16\pi}\right]_0^{L/8}$$

Step 2. At $x = \frac{L}{8}$: $\sin(\pi) = 0$. So

$$P = \frac{2}{L}\cdot\frac{L}{16} = \frac{1}{8} = 0.125$$

($\frac{L}{8}$ contains exactly one of the 8 equal half-loops, so this makes sense.)

Final answer: **(C) 0.125** ✓ Matches answer key

Question 23 Compton-shifted wavelength

23. 6 MeV X-rays are scattered off a carbon-12 target. What is the shifted wavelength λ' that will be measured at a scattering angle of 30°?

A) 5.3×10^{-13} m
B) 4.0×10^{-13} m
C) 8.6×10^{-13} m
D) 2.1×10^{-13} m

FROM THE FORMULA SHEET

$$\Delta\lambda = \frac{h}{mc}(1 - \cos\phi), \qquad E = hf, \qquad c = f\lambda$$

REMEMBER — not on the formula sheet

$\lambda = \dfrac{hc}{E}$. Use the **electron** mass in the Compton formula.

Step 1. Incident wavelength, with $E = 6\ \text{MeV} = 9.612 \times 10^{-13}$ J:

$$\lambda = \frac{(6.63 \times 10^{-34})(3.00 \times 10^{8})}{9.612 \times 10^{-13}} = 2.07 \times 10^{-13}\ \text{m}$$

Step 2. Shift:

$$\Delta\lambda = \frac{6.63 \times 10^{-34}}{(9.109 \times 10^{-31})(3.00 \times 10^{8})}(1 - \cos 30^{\circ}) = (2.426 \times 10^{-12})(0.134) = 3.25 \times 10^{-13}\ \text{m}$$

Step 3. New wavelength: $\lambda' = 2.07 \times 10^{-13} + 3.25 \times 10^{-13} = 5.3 \times 10^{-13}$ m

Final answer: **(A)** 5.3×10^{-13} **m** ✓ Matches answer key

Question 24 Ranking Compton shifts by angle

24. Rank the Compton shift that would be measured at the following angles from smallest to largest: 0º, 45º, 150º, and 210º.

A) $45^{\circ} < 0^{\circ} < 150^{\circ} < 210^{\circ}$
B) $150^{\circ} < 210^{\circ} < 0^{\circ} < 45^{\circ}$
C) $150^{\circ} = 210^{\circ} < 0^{\circ} < 45^{\circ}$
D) $0^{\circ} < 45^{\circ} < 150^{\circ} = 210^{\circ}$

FROM THE FORMULA SHEET

$\Delta\lambda \propto (1 - \cos\phi)$

$0^{\circ} : 0, \qquad 45^{\circ} : 0.293, \qquad 150^{\circ} : 1.866, \qquad 210^{\circ} : 1 - (-0.866) = 1.866$

Final answer: **(D)** $0^{\circ} < 45^{\circ} < 150^{\circ} = 210^{\circ}$ ✓ Matches answer key

Question 25 Accelerating voltage for a given wavelength

25. What potential difference would accelerate an electron such that it has a wavelength of 5.48×10^{-11} m?

A) 1.2 kV
B) 75 kV
C) 50 kV
D) 0.5 kV

FROM THE FORMULA SHEET

$$p = \frac{h}{\lambda}, \quad U = qV$$

REMEMBER — not on the formula sheet

$$K = \frac{p^2}{2m}$$

Step 1. Momentum: $p = \dfrac{6.63 \times 10^{-34}}{5.48 \times 10^{-11}} = 1.210 \times 10^{-23}$ kg·m·s^{-1}

Step 2. Kinetic energy:

$$K = \frac{p^2}{2m_e} = \frac{(1.210 \times 10^{-23})^2}{2(9.109 \times 10^{-31})} = 8.04 \times 10^{-17}\ \text{J}$$

Step 3. Voltage, from $K = eV$:

$$V = \frac{K}{e} = \frac{8.04 \times 10^{-17}}{1.602 \times 10^{-19}} = 502\ \text{V} \approx 0.5\ \text{kV}$$

Final answer: **(D) 0.5 kV** ✓ **Matches answer key**